Knowledge Base Entry — Industrial Research & Auditory Signal Processing

MSc Thesis Student, Auditory Signal Processing — WS Audiology (WSA)

Role Overview

Master's thesis conducted in collaboration with WS Audiology R&D in Erlangen, Germany, embedded at the interface between audiology research and signal processing. The project was explicitly framed as **exploratory research**, with a strong emphasis on learning, hypothesis testing, and critical evaluation of feasibility rather than product delivery. Worked largely independently with weekly academic and industrial supervision.

Research Context & Scope (NDA-Safe)

- Conducted exploratory research into **signal processing strategies aimed at reducing spectro-temporal masking** in complex auditory environments relevant to hearing aid users.
- The project initially explored biologically inspired approaches, drawing on concepts from cochlear implant research and auditory neuroscience.
- Through systematic investigation and literature grounding, identified that:
    - Certain biologically inspired strategies **do not transfer directly** from cochlear implants to hearing aids.
    - Problems that are perceptually well-defined (e.g. speech in noise) can mask **fundamental theoretical and perceptual limits**, even when reframed as engineering tasks.
- Actively engaged supervisors in reframing the project once it became clear that the original formulation aligned with a long-standing, unsolved research problem in auditory scene analysis.

Final Thesis Focus (Public & Defensible)

- Designed and evaluated **controlled auditory signal processing experiments** targeting spectro-temporal masking effects.
- Implemented MATLAB-based prototyping pipelines for audio processing and experimental evaluation.
- Modified and used a **GUI-based OLSA speech intelligibility test** to conduct structured listening experiments.
- Designed experiments, piloted protocols, and performed statistical analysis (MATLAB, JMP) to assess effects of processing parameters.
- Worked with normal-hearing participants as a controlled proxy population to isolate processing effects without introducing additional confounds.

Engineering Judgment & Research Practice

- Demonstrated the ability to **recognize when a problem formulation exceeds feasible scope**, and to articulate *why*—in terms of perception, physiology, and system-level complexity.

- Learned to distinguish between:
    - Algorithmic plausibility and **perceptual usefulness**
    - Biological inspiration and **biological equivalence**
- Developed a nuanced understanding that:
    - The auditory system's fine-grained mechanics and adaptability introduce constraints that cannot be bypassed by signal processing alone.
    - Negative or null results are not failures, but **boundary-setting knowledge** that reduces future risk and misdirected effort.
- Successfully defended a thesis with **non-trivial negative findings**, with explicit recognition from industrial supervision of the analytical depth and effort involved.

Professional Takeaways (Job-Focused)

- Gained deep respect for the **limits of abstraction** when working with biological systems.
- Strengthened skills in experimental design, critical literature analysis, and statistical reasoning.
- Learned that in applied research, **unaccounted factors and hidden assumptions** often dominate outcomes more than the visible algorithmic choices.
- Developed confidence in communicating complex, uncomfortable conclusions to both academic and industrial stakeholders.

Reusable Professional Insights (NDA-Safe)

- "My master's thesis taught me that understanding the *limits* of a system is often more important than optimizing within an assumed model."
- "In auditory signal processing, computational feasibility and perceptual usefulness can diverge sharply, especially when biological systems are involved."
- "Exploratory research trained me to recognize when a problem formulation itself—not the implementation—is the primary challenge."

Core Keyword Cluster

Auditory Signal Processing · Audiology · Hearing Aids · Spectro-Temporal Masking · Experimental Design · MATLAB · Perceptual Evaluation · Signal Processing Research · Medical Devices · Industrial R&D · Statistical Analysis